

NOTE ON A RESULT OF L. FUCHS ON ORDERED GROUPS.*¹

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Let G be an abelian group. If it admits a linear order, every non-zero element satisfies the relation $a > 0$ or $a < 0$, whence $na > 0$ or $na < 0$ for all $n = 1, 2, \dots$. Hence, G has the property (*): every non-zero element of G is of infinite order.

If P_0 is an arbitrary partial order on an abelian group G of type (*), it possesses a linear extension L . To see this, it is convenient to recall that a partial order on G is completely defined by its set N of elements $p \geq 0$. The latter has the characterizing properties: A) N is closed under addition, B) contains zero, C) contains no element along with its inverse except zero. Note that if positive multiples nx and $m(-x)$ are in N , then² mnx and $nm(-x) = -(mnx)$ are in N and $x = 0$.

If N is the non-negative set of a partial order P and neither x nor $-x$ is in N , three mutually exclusive cases may obtain: 1) some positive multiple nx is in N , 2) some positive multiple $m(-x)$ is in N , 3) no positive multiple nx nor $m(-x)$ is in N . In case 1, define N^* as the set of all elements of form $p + nx$, p in N , $n = 0, 1, 2, \dots$; in case 2, similarly but with x replaced by $-x$; in case 3, in either way. It is trivial that N^* satisfies A, B, C, and contains N properly.

The class of all sets N satisfying A, B, C, and containing the set N_0 of the original P_0 , is a partially ordered set under set inclusion; and every linearly ordered subclass has an upper bound, namely its set union. Hence there are maximal sets containing N_0 . But a maximal set must contain either x or $-x$ for every x of G , and the corresponding order is a linear extension of P_0 .

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¹ L. Fuchs, "On the extension of the partial order of groups," *American Journal of Mathematics*, vol. 72 (1950), pp. 191-194.

² Remark by L. Fuchs simplifying original argument.